PROJECT PROPOSAL

Project Title: Student Academic Performance Prediction System

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Abstract -

This project proposes the development of a machine learning based system to predict student's final academic performance using internal academic and engagement data. The system will help the university identify students who may require additional academic support before final examinations. The project focuses not only on building a predictive model but also on designing a deployable pipeline, a user dashboard and a maintenance plan for periodic model updates as new student data becomes available.

Problem Statement -

In many universities, students who struggle academically are often identified only after final exam results are declared, which leaves little time for effective intervention. There is a need for an early prediction system that can help faculty recognize at-risk students during the semester. **We aim to predict a student's final academic performance using attendance records, internal assessment scores and engagement indicators so that the university can identify students who may need additional academic support and take timely intervention measures.** This prediction will help the university provide timely mentoring, improve student outcomes and reduce academic failures.

Proposed Solution -

The proposed system will use historical student data to train machine learning models capable of predicting final performance categories (e.g., High, Medium, Low). The solution will include:

- A structured data pipeline connected to a SQL database
- Training and evaluation of traditional machine learning models
- A prediction-ready system for new student inputs
- A dashboard for analysis and visualization
- A model lifecycle plan for periodic retraining and updates

The system is designed to be scalable and maintainable, reflecting real-world deployment practices.

Data Description -

The dataset will contain structured academic records of university students. The data will be synthetically generated using a custom Python data generation script designed to simulate realistic student performance patterns. This script will be included in the project repository instead of uploading large raw datasets.

Each student record will include the following features:

- **Attendance Percentage** – Overall attendance during the semester
- **Internal Test Scores** – Average marks obtained in internal assessments
- **Assignment Scores** – Average marks obtained in coursework and assignments
- **Number of Backlogs** – Count of previously failed subjects, if any
- **Engagement Score** – A numerical indicator representing class participation and academic involvement

The target variable will be **Final Academic Performance**, categorized into performance levels such as High, Medium and Low based on academic outcomes.

New student data will be generated and added to the dataset at the end of every academic term. This periodic data expansion will be used to retrain and update the model, simulating real-world data growth and supporting the model lifecycle requirement.

Machine learning models -

Up to three traditional machine learning models will be trained and compared. Possible models include:

- Logistic Regression
- Decision Tree Classifier
- Random Forest Classifier

These models will be evaluated using standard performance metrics such as accuracy, precision, recall and F1-score. The goal is to understand model behaviour and performance rather than achieve maximum accuracy.

Dashboard and user interface -

A dashboard will be developed to present:

- Visual analysis of student performance trends
- Model performance metrics
- Input fields for new student data
- Real-time prediction outputs

The dashboard will act as an interface between users (faculty/administrators) and the prediction system.

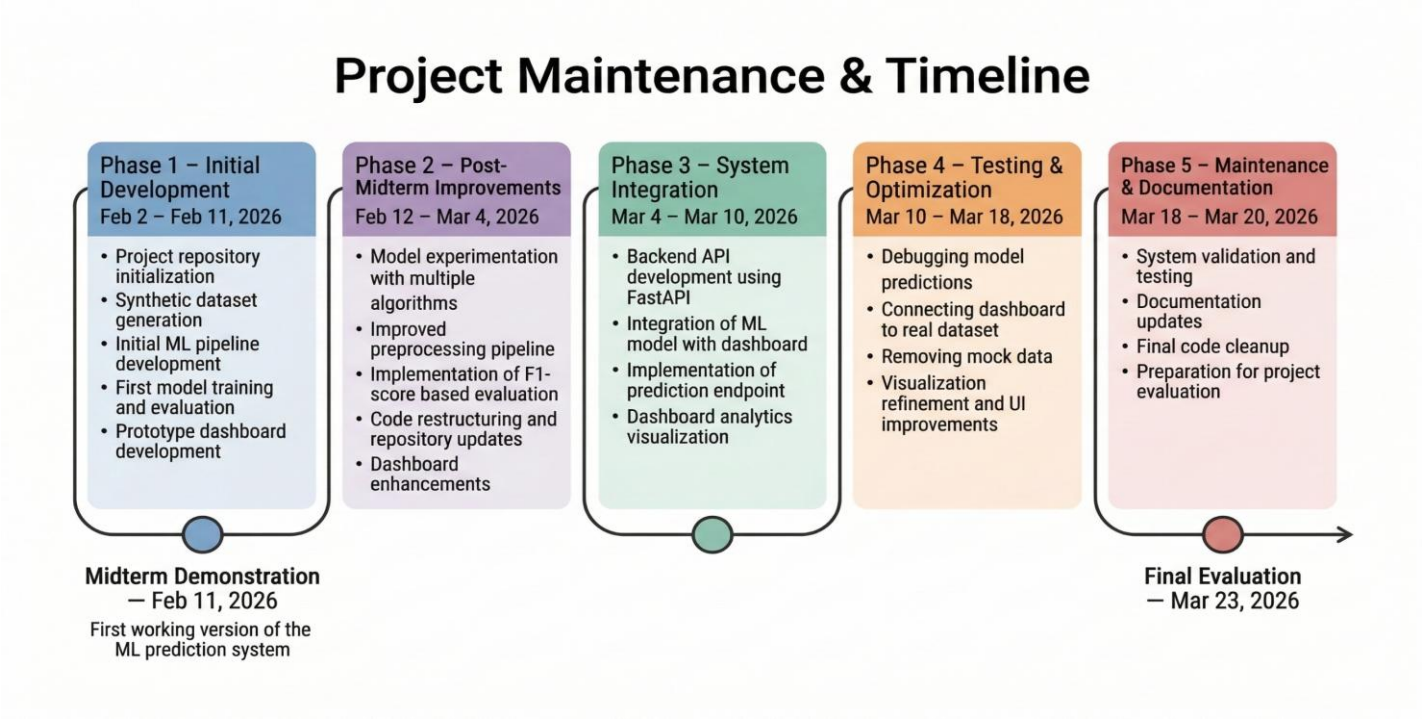
Prediction Capability -

The trained model will be integrated into the dashboard so that users can enter new student details and receive predicted performance outcomes. This ensures the model is usable in a real academic setting and not limited to offline analysis.

Model improvement and maintenance plan -

The system will follow a model lifecycle approach. At regular intervals (e.g., each semester), new student data will be added to the database. The model will then be retrained using the updated dataset. Older model versions will be stored to maintain version history and the dashboard will always use the most recent model.

Project Timeline -



Expected Output -

The final system is expected to provide:

- Predicted student performance categories
- Visual insights into academic trends
- A usable prediction interface for university stakeholders
- A maintainable ML pipeline that supports periodic updates

Version control and repository plan -

The project will be maintained in a public GitHub repository with clear version control practices. The repository will include Python scripts, database connection logic, model training, retraining scripts and dashboard code. Large datasets and model files will not be uploaded; instead, data generation scripts and instructions will be provided.